

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
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Date:	03/08/2026	Duration:	60 minutes

Code of Conduct

I Dhanushree S-192421218 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Dhanushree S

Annexure A – Scenario-Based Requirement Analysis

AI-Based Crop Disease Detection and Farmer Advisory Platform

AI-Based Crop Disease Detection and Farmer Advisory Platform

1. Introduction

Agriculture is one of the most important sectors contributing to the economic development of many countries, particularly India. A large percentage of the population depends on farming as their primary source of income. However, crop diseases remain one of the biggest challenges faced by farmers. These diseases significantly reduce crop quality and yield, resulting in financial losses and food insecurity. Early detection of crop diseases is essential for effective treatment and improved agricultural productivity.

Traditionally, farmers identify crop diseases through manual inspection or by consulting agricultural experts. This approach is often time-consuming, expensive, and unavailable in many rural areas where expert assistance is limited. Farmers may also misidentify diseases due to similar symptoms, leading to incorrect pesticide usage and further damage to crops.

Recent advancements in Artificial Intelligence (AI), Machine Learning (ML), and Computer Vision have made it possible to automate crop disease detection using image analysis. AI models trained on large datasets can accurately identify crop diseases from leaf images and provide instant recommendations for treatment and prevention.

The **AI-Based Crop Disease Detection and Farmer Advisory Platform** is a smart agricultural software solution designed to assist farmers in detecting crop diseases through uploaded crop images. The platform analyzes the images using AI algorithms and provides disease identification, confidence scores, fertilizer recommendations, pesticide suggestions, weather-based advisories, and preventive farming practices. This system aims to support farmers in making informed decisions, improving crop productivity, reducing crop losses, and promoting sustainable agriculture.

2. Scenario Analysis

The proposed platform addresses the increasing demand for smart farming solutions by combining Artificial Intelligence with digital agricultural services. Farmers can access the platform through a web application or mobile application. After logging into the system, users upload images of diseased crops using their smartphone camera or gallery.

The uploaded image is processed by the AI prediction engine, which compares the image with trained disease datasets. The system identifies the disease and displays the prediction result along with the confidence percentage. Based on the diagnosis, the platform recommends suitable pesticides, fertilizers, organic treatments, irrigation advice, and preventive measures.

The platform also integrates weather forecasting services to provide climate-based recommendations. Agricultural experts can review AI predictions, update treatment recommendations, and improve the knowledge base. Administrators manage users, crop databases, reports, and overall system maintenance.

The proposed system significantly reduces the dependency on manual disease diagnosis while improving accessibility to agricultural expertise.

3. Business Domain

The proposed software belongs to the **Agricultural Technology (AgriTech)** domain. AgriTech integrates advanced technologies such as Artificial Intelligence, Machine Learning, Computer Vision, Internet of Things (IoT), Cloud Computing, and Data Analytics to modernize traditional farming practices.

The primary objective of the AgriTech domain is to improve farming efficiency, enhance crop productivity, reduce operational costs, and support sustainable agriculture through digital transformation.

This project specifically focuses on:

- Crop disease detection using AI
- Digital farmer advisory services
- Precision agriculture
- Smart farming technologies
- Decision support systems
- Agricultural data analytics

4. Existing Challenges

Farmers currently face several challenges that affect agricultural productivity and profitability. The major challenges are:

4.1 Manual Disease Identification

Farmers rely on visual observation to identify diseases. Similar symptoms among different diseases often result in incorrect diagnosis.

4.2 Lack of Agricultural Experts

Many rural areas do not have immediate access to agricultural officers or plant pathologists, causing delays in obtaining professional advice.

4.3 Delayed Disease Detection

Diseases are often detected only after they have spread significantly, reducing the effectiveness of treatment.

4.4 Excessive Pesticide Usage

Due to uncertainty in disease identification, farmers frequently use inappropriate or excessive pesticides, increasing production costs and causing environmental pollution.

4.5 Low Crop Yield

Untreated diseases reduce crop quality and quantity, affecting farmers' income and food production.

4.6 Weather Uncertainty

Changing weather conditions influence disease spread and farming decisions. Farmers often lack timely weather-based guidance.

4.7 Limited Digital Farming Support

Many farmers still depend on traditional farming methods and lack access to modern digital advisory systems.

4.8 Poor Record Management

Farmers rarely maintain historical records of diseases, treatments, and crop performance, making future decision-making difficult.

5. Proposed Solution

The AI-Based Crop Disease Detection and Farmer Advisory Platform addresses these challenges by providing an intelligent digital solution.

The proposed platform allows farmers to upload crop images using a mobile device or web application. The AI model analyzes the uploaded image and predicts the crop disease with high accuracy. After disease detection, the platform recommends:

- Suitable pesticides
- Fertilizer recommendations
- Organic treatment methods
- Disease prevention techniques
- Weather-based farming advisory
- Crop management suggestions

The system also stores previous disease reports, enabling farmers to monitor crop health over time. Agricultural experts can review AI predictions and continuously improve the recommendation database.

6. Project Objectives

The main objectives of the proposed system are:

- Develop an AI-powered crop disease detection platform.
- Detect crop diseases using image processing and deep learning algorithms.
- Improve the accuracy of disease diagnosis.
- Provide instant treatment recommendations.
- Recommend appropriate fertilizers and pesticides.
- Provide weather-based agricultural advisory.
- Reduce crop losses caused by late disease detection.
- Improve agricultural productivity.
- Support sustainable farming practices.
- Maintain historical disease records for future analysis.

- Enable agricultural experts to validate disease predictions.
- Provide a user-friendly interface for farmers.
- Support multilingual communication for rural users.
- Generate detailed crop health reports.
- Improve overall decision-making in agriculture.

7. Project Vision

The vision of the AI-Based Crop Disease Detection and Farmer Advisory Platform is to transform traditional farming into smart agriculture by leveraging Artificial Intelligence and digital technologies. The platform aims to provide farmers with quick, accurate, and affordable disease diagnosis while promoting sustainable farming practices. It envisions creating a technology-driven agricultural ecosystem where farmers can easily access expert knowledge, reduce crop losses, improve productivity, and contribute to national food security.

8. Stakeholders

The successful implementation of the system involves multiple stakeholders.

Stakeholder	Role	Responsibilities
Farmer	Primary User	Upload crop images, receive disease diagnosis, follow recommendations
Agricultural Expert	Domain Expert	Validate AI predictions, update treatment guidelines
Administrator	System Administrator	Manage users, reports, crop database, and system security
AI Prediction Engine	Intelligent Component	Analyze crop images and detect diseases
Government Agriculture Department	External Stakeholder	Provide agricultural policies, awareness programs, and support schemes
Researchers	Data Scientists	Improve AI model accuracy and update datasets

Development Team	Software Engineers	Design, develop, test, and maintain the platform
Product Owner	Project Owner	Define business requirements and prioritize features
Scrum Master	Agile Facilitator	Coordinate Agile activities and sprint planning

9. System Scope

The scope of the AI-Based Crop Disease Detection and Farmer Advisory Platform includes the following functionalities:

In Scope

- Farmer Registration and Login
- Crop Image Upload
- AI-Based Disease Detection
- Disease Prediction with Confidence Score
- Fertilizer Recommendation
- Pesticide Recommendation
- Organic Treatment Suggestions
- Weather-Based Farming Advisory
- Crop Health History
- Expert Verification
- Report Generation
- Notification System
- Admin Dashboard
- User Management
- Disease Database Management

Out of Scope

- Online purchase of pesticides and fertilizers
- Drone-based crop monitoring

- Soil sensor integration
- Automatic irrigation control
- Financial loan processing
- Crop insurance management

10. Expected Benefits

The proposed platform provides numerous benefits to farmers and agricultural organizations:

- Early detection of crop diseases.
- Increased crop productivity.
- Reduced crop losses.
- Improved disease diagnosis accuracy.
- Reduced unnecessary pesticide usage.
- Faster access to agricultural expertise.
- Better decision-making using AI.
- Increased farmer awareness.
- Enhanced food security.
- Promotion of sustainable agriculture.
- Digital transformation of farming practices.
- Easy access through smartphones and web applications.

Functional and Non-Functional Requirements

11. Functional Requirements

Functional requirements describe the specific functions and services that the AI-Based Crop Disease Detection and Farmer Advisory Platform must provide to meet user needs. These requirements define the behavior of the system and the interactions between users and the application.

JFR1 – User Registration

Description:

The system shall allow new farmers to register by providing personal information such as name, email, mobile number, password, and location.

Inputs:

- Name
- Email Address
- Mobile Number
- Password
- Village/Location

Outputs:

- Successful registration confirmation.
- User account creation.

Priority: High

FR2 – User Login

Description:

The system shall authenticate registered users using their email or mobile number and password.

Inputs:

- Username/Email
- Password

Outputs:

- Successful login.
- Error message for invalid credentials.

Priority: High

FR3 – Password Recovery

Description:

The system shall allow users to reset forgotten passwords using email or OTP verification.

Inputs:

- Registered Email or Mobile Number

Outputs:

- Password reset link or OTP.

Priority: Medium

FR4 – Farmer Profile Management

Description:

Farmers shall be able to update their profile information such as address, crop details, preferred language, and contact information.

Priority: Medium

FR5 – Crop Image Upload

Description:

The system shall allow farmers to upload crop images from their mobile camera or gallery.

Inputs:

- Crop Image

Outputs:

- Image uploaded successfully.

Priority: High

FR6 – Image Processing

Description:

The uploaded crop image shall be preprocessed before AI analysis by resizing, noise removal, normalization, and feature extraction.

Priority: High

FR7 – AI-Based Disease Detection

Description:

The AI model shall analyze the uploaded crop image and predict the disease using machine learning and computer vision techniques.

Outputs:

- Disease Name
- Confidence Score
- Disease Category

Priority: High

FR8 – Disease Information

Description:

The system shall display complete information about the detected disease, including symptoms, causes, and prevention methods.

Priority: High

FR9 – Treatment Recommendation

Description:

The system shall recommend appropriate treatment methods based on the detected disease.

Recommendations Include:

- Chemical Treatment
- Organic Treatment
- Preventive Measures

Priority: High

FR10 – Fertilizer Recommendation

Description:

The system shall recommend suitable fertilizers according to the crop type and detected disease.

Priority: Medium

FR11 – Pesticide Recommendation

Description:

The system shall recommend appropriate pesticides with dosage instructions and safety precautions.

Priority: High

FR12 – Weather Advisory

Description:

The system shall display weather forecasts and provide farming recommendations based on current weather conditions.

Information Includes:

- Temperature
- Humidity
- Rainfall
- Wind Speed

Priority: High

FR13 – Disease History

Description:

The system shall maintain previous disease detection records for each farmer.

Priority: Medium

FR14 – Report Generation

Description:

The system shall generate downloadable reports containing disease diagnosis, recommendations, and treatment details.

Report Format:

- PDF
- Excel

Priority: Medium

FR15 – Notification System

Description:

The system shall send notifications regarding weather alerts, disease outbreaks, treatment reminders, and government schemes.

Priority: Medium

FR16 – Agricultural Expert Module

Description:

Experts shall review AI predictions and update treatment recommendations if necessary.

Priority: Medium

FR17 – Admin Dashboard

Description:

The administrator shall manage users, disease databases, AI models, reports, and system settings.

Priority: High

FR18 – Search Functionality

Description:

Users shall search diseases by crop name or disease name.

Priority: Low

FR19 – Multilingual Support

Description:

The system shall support multiple regional languages such as English, Tamil, Hindi, and Telugu.

Priority: Medium

FR20 – Feedback Module

Description:

Farmers shall submit feedback regarding AI prediction accuracy and advisory services.

Priority: Low

12. Non-Functional Requirements

Non-functional requirements describe the quality attributes and operational characteristics of the proposed system.

12.1 Performance Requirements

- Disease prediction should be completed within **5 seconds**.
- Image upload should take less than **3 seconds** under normal network conditions.
- The system should support more than **10,000 concurrent users**.
- Dashboard loading time should be less than **2 seconds**.
- Reports should be generated within **5 seconds**.

12.2 Security Requirements

- Passwords shall be encrypted using secure hashing algorithms.
- Secure user authentication shall be implemented.
- Role-based access control shall restrict unauthorized access.
- All communications shall use HTTPS encryption.
- User data shall be protected against unauthorized access.
- Regular security updates shall be performed.

12.3 Reliability Requirements

- System availability should be at least **99%**.
- Automatic database backup shall be performed daily.
- The platform shall recover from failures without data loss.
- AI prediction services should remain stable during heavy usage.

12.4 Usability Requirements

- The user interface shall be simple and intuitive.
- The application shall be easy for farmers with basic smartphone knowledge.
- Important buttons shall be clearly labeled.
- Navigation shall be simple and user-friendly.
- Multiple language support shall improve accessibility.

12.5 Scalability Requirements

- The platform shall support additional crops and diseases.
- New AI models shall be integrated without redesigning the system.
- The database shall accommodate increasing user data efficiently.

- Cloud deployment shall support horizontal scaling.

12.6 Maintainability Requirements

- Modular software architecture shall simplify maintenance.
- Source code shall follow coding standards.
- Documentation shall be updated regularly.
- Bugs shall be resolved with minimal downtime.
- Future features shall be integrated easily.

12.7 Availability Requirements

- The platform shall be available **24×7**.
- Scheduled maintenance shall occur during low-traffic periods.
- Backup servers shall ensure uninterrupted service.

12.8 Compatibility Requirements

The platform shall support:

- Android Smartphones
- iOS Devices
- Windows PCs
- Linux Systems
- macOS
- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Safari Browser

12.9 Portability Requirements

The software shall operate on:

- Mobile Application
- Web Application

- Cloud Infrastructure

12.10 Data Integrity Requirements

- Uploaded crop images shall remain unchanged.
- Prediction records shall not be modified without authorization.
- Reports shall be stored securely.
- Duplicate records shall be prevented.

12.11 Privacy Requirements

- Farmer personal information shall remain confidential.
- Images shall only be used for disease analysis.
- User consent shall be obtained before collecting data.
- Privacy regulations shall be followed.

12.12 Backup and Recovery Requirements

- Daily database backup.
- Weekly full system backup.
- Disaster recovery procedures shall be implemented.
- Data restoration shall be possible within a reasonable time.

Summary of Requirements

Functional Requirements

- User Registration
- Login
- Password Recovery
- Profile Management
- Crop Image Upload
- Image Processing
- AI Disease Detection
- Disease Information
- Treatment Recommendation

- Fertilizer Recommendation
- Pesticide Recommendation
- Weather Advisory
- Disease History
- Report Generation
- Notifications
- Agricultural Expert Module
- Admin Dashboard
- Search Functionality
- Multilingual Support
- Feedback Module

Non-Functional Requirements

- Performance
- Security
- Reliability
- Usability
- Scalability
- Maintainability
- Availability
- Compatibility
- Portability
- Data Integrity
- Privacy
- Backup and Recovery

Identify Actors and Use Cases

3. 1 Actors

Actors are the users or external systems that interact with the AI-Based Crop Disease Detection and Farmer Advisory Platform.

Primary Actors

Farmer

The farmer is the main user of the system. Farmers upload crop images, receive disease predictions, view treatment recommendations, and access weather-based farming advice.

Agricultural Expert

Agricultural experts verify AI-generated disease predictions, update treatment recommendations, and provide expert guidance to improve the accuracy of the advisory system.

Administrator

The administrator manages users, crop databases, disease information, reports, AI models, and overall system maintenance.

Secondary Actors

AI Prediction Engine

Processes uploaded crop images using Artificial Intelligence and Machine Learning algorithms to detect diseases.

Weather API

Provides real-time weather forecasts used for generating weather-based farming recommendations.

Database Server

Stores farmer details, crop images, disease history, reports, and recommendation data.

Notification Service

Sends SMS, email, or mobile notifications regarding disease alerts, weather updates, and farming recommendations.

3.2 Major Use Cases

Farmer

- Register Account
- Login
- Manage Profile
- Upload Crop Image
- Capture Image Using Camera
- View Disease Prediction
- View Confidence Score
- Receive Treatment Recommendation
- Receive Fertilizer Recommendation
- Receive Pesticide Recommendation
- View Weather Advisory
- View Disease History
- Download Reports
- Give Feedback
- Logout

Agricultural Expert

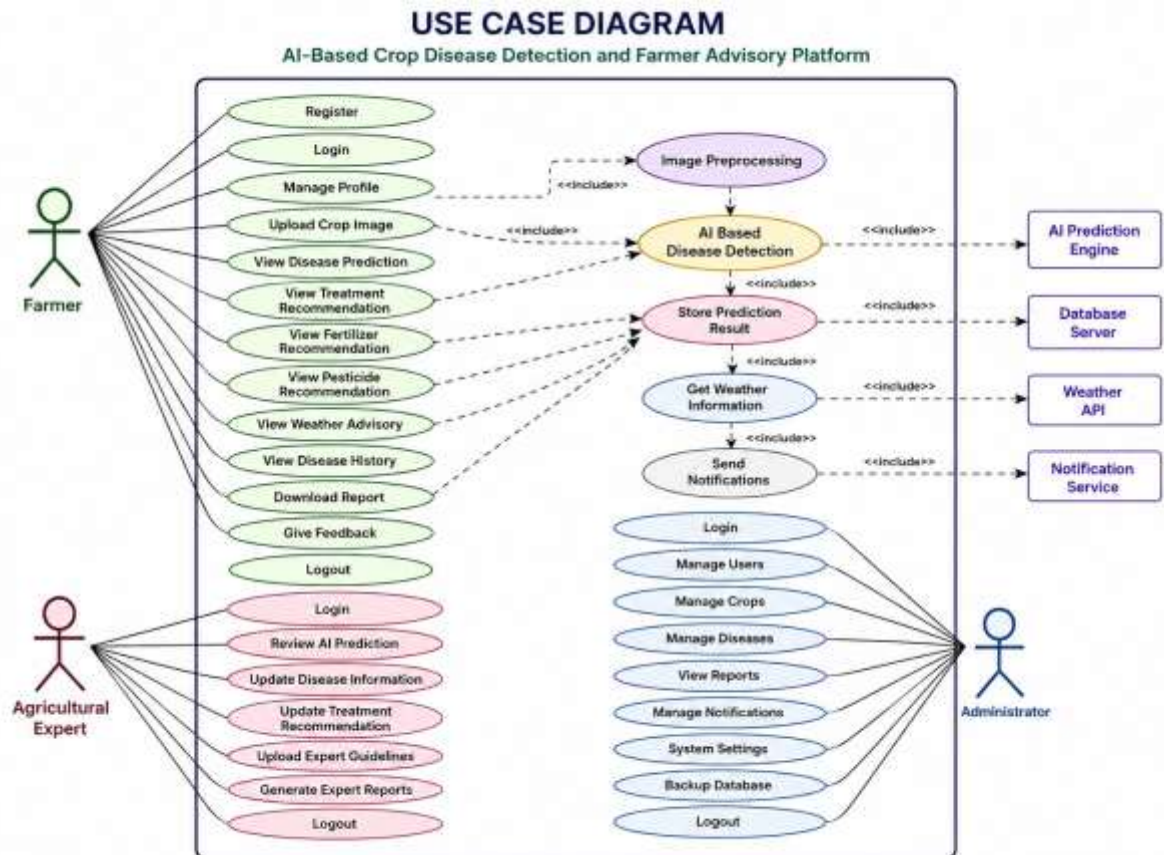
- Login
- Review AI Prediction
- Update Disease Information
- Modify Treatment Recommendation
- Upload Expert Guidelines
- Generate Expert Reports

Administrator

- Login
- Manage Farmers
- Manage Experts
- Manage Crop Database
- Manage Disease Database

- Monitor AI Accuracy
- Generate Reports
- Manage Notifications
- System Maintenance
- Backup Database

3.3 Use Case Diagram (Text Representation)



3.4 Use Case Descriptions

Use Case 1 – Upload Crop Image

Actor: Farmer

Purpose: Upload crop image for disease detection.

Precondition

The farmer must be logged into the system.

Main Flow

1. Farmer opens Upload Image page.
2. Selects image from gallery or camera.
3. Uploads image.
4. System validates image.
5. AI processes the image.

Postcondition

Image is successfully stored and analyzed.

Use Case 2 – Disease Detection

Actor

AI Prediction Engine

Purpose

Identify disease using AI.

Main Flow

1. Receive uploaded image.
2. Process image.
3. Detect disease.
4. Calculate confidence score.
5. Display result.

Postcondition

Disease report generated.

Use Case 3 – Farmer Advisory

Actor

Farmer

Purpose

Receive recommendations.

Main Flow

1. Open prediction result.
2. View treatment.
3. View fertilizer.
4. View pesticide.
5. View weather advice.

Postcondition

Farmer receives complete advisory.

Use Case 4 – Expert Review

Actor

Agricultural Expert

Purpose

Improve AI recommendation.

Main Flow

1. Login.
2. Review prediction.
3. Modify recommendation.
4. Save changes.

Use Case 5 – Report Generation

Actor

Administrator

Purpose

Generate reports.

Main Flow

1. Select report type.
2. Choose date.
3. Generate report.
4. Download PDF.

Software Requirement Specification (SRS)

4.1 Introduction

The AI-Based Crop Disease Detection and Farmer Advisory Platform is an intelligent agricultural software system developed to assist farmers in detecting crop diseases using Artificial Intelligence and Computer Vision. The platform provides timely disease diagnosis, treatment recommendations, fertilizer suggestions, weather-based advisory, and expert guidance to improve crop health and agricultural productivity.

4.2 Problem Description

Farmers often struggle to identify crop diseases accurately due to limited access to agricultural experts, delayed diagnosis, and insufficient technical knowledge. These issues lead to crop damage, excessive pesticide usage, and financial losses. The proposed system solves these problems by providing an AI-powered digital platform capable of detecting diseases quickly and recommending appropriate solutions.

4.3 Scope of the System

The scope of the system includes:

- Farmer Registration and Login
- Crop Image Upload
- AI-Based Disease Detection
- Treatment Recommendation
- Fertilizer Recommendation
- Pesticide Recommendation
- Weather Advisory
- Disease History
- Expert Review
- Admin Dashboard
- Report Generation
- Notification System

4.4 Functional Requirements

The functional requirements include:

- User Registration
- User Login
- Password Recovery
- Crop Image Upload
- AI Disease Detection
- Disease Prediction
- Treatment Recommendation
- Fertilizer Recommendation
- Pesticide Recommendation
- Weather Advisory
- Report Generation
- Disease History

- Notification Management
- Expert Verification
- Admin Dashboard

4.5 Non-Functional Requirements

The system should satisfy the following quality requirements:

- High Performance
- Strong Security
- High Reliability
- Easy Usability
- High Availability
- Scalability
- Maintainability
- Compatibility
- Data Integrity
- Privacy Protection

4.6 Data Requirements

The platform stores the following information:

Farmer Data

- Farmer ID
- Name
- Email
- Mobile Number
- Address

Crop Data

- Crop Name
- Crop Variety
- Crop Image

Disease Data

- Disease Name
- Symptoms
- Treatment
- Fertilizer
- Pesticide

Prediction Data

- Prediction ID
- Prediction Result
- Confidence Score
- Prediction Date

Weather Data

- Temperature
- Humidity
- Rainfall
- Wind Speed

4.7 External Interface Requirements

User Interface

- Mobile Application
- Web Application

Hardware Interface

- Smartphone Camera
- Computer

Software Interface

- TensorFlow

- OpenCV
- Flask
- MySQL
- REST API

Communication Interface

- Internet
- Cloud Server
- Weather API

4.8 System Features

The proposed platform includes the following features:

- Secure User Registration
- Secure Login
- Crop Image Upload
- AI Disease Detection
- Confidence Score Prediction
- Treatment Recommendation
- Fertilizer Recommendation
- Pesticide Recommendation
- Weather Advisory
- Disease History
- Expert Verification
- Notification System
- Report Generation
- Admin Dashboard
- Multilingual Support

5. Assumptions and Constraints

Assumptions

- Farmers have smartphones with internet connectivity.
- Uploaded crop images are clear and of good quality.
- AI models are trained using reliable datasets.
- Weather API provides accurate weather information.
- Agricultural experts regularly update treatment recommendations.

Constraints

Technical Constraints

- Requires internet connection.
- Requires cloud storage.
- AI model requires periodic updates.

Operational Constraints

- Farmers need basic knowledge of smartphone usage.
- Availability of agricultural experts.

Economic Constraints

- Cloud hosting cost.
- AI model maintenance cost.
- Database storage cost.

Legal Constraints

- Compliance with data privacy regulations.
- Secure storage of farmer information.

Time Constraints

- Total Project Duration: **6 Months**
- Sprint Duration: **2 Weeks**

6. Requirement Validation

Completeness

All stakeholder requirements have been identified and documented.

Consistency

Functional and non-functional requirements are consistent with the project objectives.

Ambiguity Check

Requirements are written in simple, clear, and measurable language to avoid ambiguity.

Redundancy Check

Duplicate requirements have been eliminated.

Missing Requirement Analysis

Future enhancements may include:

- Drone-based crop monitoring
- IoT soil sensor integration
- Voice assistant in regional languages
- Offline AI prediction
- Blockchain-based agricultural records

Suggestions for Improvement

- Improve AI model accuracy using larger datasets.
- Add multilingual voice assistance.
- Integrate IoT sensors for soil monitoring.
- Include government subsidy notifications.
- Develop an offline prediction mode.
- Add chatbot support for farmers.

Conclusion

The **AI-Based Crop Disease Detection and Farmer Advisory Platform** is a modern agricultural solution that combines Artificial Intelligence, Machine Learning, and Computer Vision to assist farmers in early disease detection and decision-making. The Software Requirement Specification clearly defines the project scope, stakeholders, functional and non-functional requirements, use cases, data requirements, external interfaces, assumptions, constraints, and validation process. This system will improve crop productivity, reduce agricultural losses, promote sustainable farming, and provide reliable digital advisory services. It also establishes a strong foundation for future enhancements such as IoT integration, drone monitoring, and predictive analytics, making it a scalable and efficient solution for smart agriculture.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.

Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.
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Criteria	Mark
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25